

1 Solution — GIAM 7.2

Problem 5

1.1 Problem

The five tetrominoes (familiar to players of the video game *Tetris*) are relatives of dominoes made up of four small squares. All together these five tetrominoes contain **20** squares, so it is conceivable that they could be used to tile a 4×5 chessboard. Prove that this is actually impossible.

1.2 Setup

A *tiling* uses each of the five tetrominoes exactly once (rotations and reflections allowed), with no overlap and no uncovered cells. The five tetrominoes are shown below — by convention I label them **I, T, S, L, J** — and total $4 + 4 + 4 + 4 + 4 = 20$ cells, the same as the 4×5 board. So a tiling is at least *numerically* plausible. The argument below shows that no tiling exists, by way of a coloring (parity) invariant.

1.3 The Coloring Argument

Color the 4×5 board with the standard chess pattern: cell (i, j) is **dark** if $i + j$ is even, **light** if $i + j$ is odd. (Either choice of which parity is dark works the same way — we just need *some* fixed two-coloring.)

4 × 5 chessboard: 10 dark + 10 light

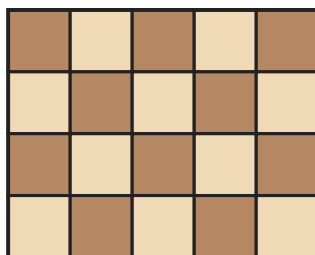


Figure 1.1: The 4×5 board under the standard chess coloring contains exactly 10 dark squares and 10 light squares.

The board has $4 \cdot 5 = 20$ cells split evenly: 10 **dark**, 10 **light**. Any tiling must cover *exactly* this many of each color, just as in the mutilated-chessboard argument (Problem 3).

1.3.1 Each Tetromino's Dark/Light Signature

The key observation is that the five tetrominoes do *not* all behave the same way under the chess coloring.

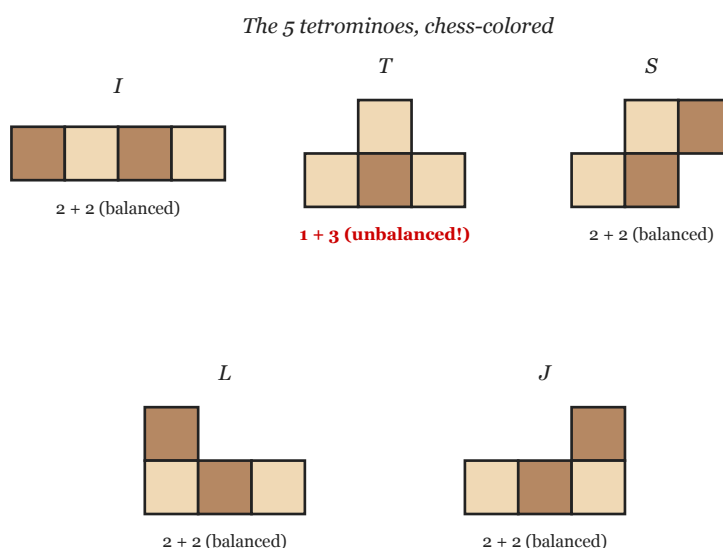


Figure 1.2: Chess-colored views of the five tetrominoes I, T, S, L, J. The I, S, L, J tetrominoes always cover 2 dark + 2 light cells (in any orientation); the T tetromino is the unique offender — it always covers 3 of one color and 1 of the other.

Claim. In *any* placement (translation, rotation, reflection) on the chess-colored grid:

- *I, S, L, J* each cover exactly 2 dark + 2 light cells.
- *T* covers either 3 dark + 1 light, *or* 1 dark + 3 light — never 2 + 2.

Why for I, S, L, J. Each of these four tetrominoes admits a *partition into two dominoes*. (E.g., the *I*-tetromino is a 1×4 bar, splittable into two 1×2 dominoes; the *L*-tetromino splits as $(0, 0)-(1, 0)$ plus $(1, 1)-(1, 2)$; analogously for *S* and *J*.) Each domino, by adjacency, covers one cell of each color, so two dominoes cover 2 + 2.

Why for T. The *T*-tetromino has a distinguished “center” cell with three neighbors inside the piece (the cell where the stem meets the bar). All three neighbors of this center sit at chess-coloring parity *opposite* to the center. So the four cells of *T* split as

$$\underbrace{1}_{\text{center}} + \underbrace{3}_{\text{three neighbors of opposite color}} .$$

Equivalently, T is *not* tileable by two dominoes — any attempted split leaves one pair non-adjacent — which is precisely the obstruction to having a $2 + 2$ chess signature.

1.4 Adding Up the Signatures

A complete tiling of the 4×5 board with all five tetrominoes must cover *exactly* 10 dark cells in total. Let d_X denote the number of dark cells covered by tetromino X in a hypothetical tiling. Then

$$d_I + d_T + d_S + d_L + d_J = 10.$$

The claim tells us:

$$d_I = 2, \quad d_S = 2, \quad d_L = 2, \quad d_J = 2, \quad d_T \in \{1, 3\}.$$

Substituting,

$$2 + d_T + 2 + 2 + 2 = 10 \iff d_T = 2.$$

But $d_T \in \{1, 3\}$, so $d_T \neq 2$. Contradiction.

Hence no tiling of the 4×5 board by the five tetrominoes exists. ■

1.5 Concrete Arithmetic

To restate without unknowns: any tiling would cover

$$\underbrace{4 \cdot (2 + 2)}_{\text{from } I, S, L, J} + \underbrace{(3 + 1) \text{ or } (1 + 3)}_{\text{from } T} = (11 + 9) \text{ or } (9 + 11)$$

dark + light cells. Neither matches the board's $10 + 10$ — the T 's unavoidable imbalance shifts the total by ± 1 off the required $10 + 10$, regardless of how T is oriented.

1.6 Remark — Why the T Is Special

Among the seven *one-sided* tetrominoes (I, O, T, L, J, S, Z), the T is the **unique** one with a $3 + 1$ chess-coloring signature; all others have $2 + 2$. The reason is structural: a

tetromino has a $2 + 2$ signature iff it admits a perfect domino tiling, and only the T fails this test.

1.7 Remark — A Family of Parity Arguments

This is the third coloring/parity proof in this section’s exercises:

- **Problem 3** (mutilated chessboard): removing two same-color corners breaks the dark/light balance, foiling any domino tiling.
- **Problem 4** (odd-side board, corner-color removed): removing the *unique extra* dark square restores balance and a tiling exists (constructed explicitly).
- **Problem 5** (this one): the five tetrominoes’ combined signature is $11 + 9$ rather than $10 + 10$, so a 4×5 tiling fails.

In each case, a clever two-coloring of cells turns a geometric question (does a tiling exist?) into an arithmetic one (do the color counts match?). The T -tetromino’s $3 + 1$ signature is the smallest, most elegant example of an “obstruction tetromino” — a piece whose color signature alone rules out certain placements.

1.8 Remark — The Result Is Sharp

The result depends on having *exactly one* T -tetromino among the five. If instead we used:

- *Two* T -tetrominoes (with appropriate orientations giving $3 + 1$ and $1 + 3$), the imbalances cancel: $2 \cdot (3 + 1) + 3 \cdot (2 + 2)$ as $(3 + 1) + (1 + 3) + 3 \cdot (2 + 2) = 10 + 10$. The arithmetic works, so the obstruction vanishes; whether such tilings *actually* exist becomes a separate geometric question.
- *Zero* T -tetrominoes (e.g., the I, O, L, J, S set — substituting O for T): $5 \cdot (2 + 2) = 10 + 10$. Arithmetic again checks out, and in fact this set *does* admit several tilings of the 4×5 board.

So this argument exactly identifies the parity obstruction; tilings beyond this obstruction are governed by finer constraints (corner geometry, edge contacts, etc.).